

What does it mean to test a hypothesis?

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Key Points

Statistical inference (e.g. Hypothesis tests and confidence intervals) is **inference** — reasoning about the unobserved.

p -values require probability distributions.

Randomization (or Design) + a Hypothesis + a Test Statistic Function can provide probability distributions representing the hypothesis (and thus p -values).

Counterfactual Causal Inference

How can we use what we **see** to learn about what we want to **know** ?

City	Pair	Treatment	Turnout		Newspaper	y_1	y_0
			Baseline	Outcome			
Saginaw	1	0	17	16		?	16
Sioux City	1	1	21	22	Sioux City Journal	22	?
Battle Creek	2	0	13	14		?	14
Midland	2	1	12	7	Midland Daily News	7	?
Oxford	3	0	26	23		?	23
Lowell	3	1	25	27	Lowell Sun	27	?
Yakima	4	0	48	58		?	58
Richland	4	1	41	61	Tri-City Herald	61	?

Design and outcomes in the Newspapers Experiment. The Treatment column shows treatment randomized within pair with the newspaper ads as 1 and lack of treatment as 0. The potential outcomes are y_1 for treatment and y_0 for control. [Panagopoulos \(2006\)](#) provides more detail on the design of the experiment.

Counterfactual Causal Inference

How can we use what we see to learn about potential outcomes?

City	Pair	Treatment	Turnout		Newspaper	y_1	y_0
			Baseline	Outcome			
Saginaw	1	0	17	16		?	16
Sioux City	1	1	21	22	Sioux City Journal	22	?
Battle Creek	2	0	13	14		?	14
Midland	2	1	12	7	Midland Daily News	7	?
Oxford	3	0	26	23		?	23
Lowell	3	1	25	27	Lowell Sun	27	?
Yakima	4	0	48	58		?	58
Richland	4	1	41	61	Tri-City Herald	61	?

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Counterfactual Causal Inference

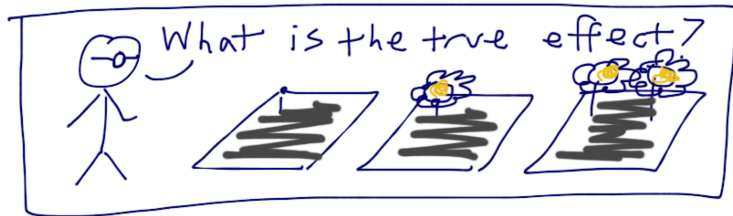
How can we use what we **see** to learn about a causal effect $\tau_i = f(y_{i,1}, y_{i,0})$?

City	Pair	Treatment	Turnout		Newspaper	y_1	y_0
			Baseline	Outcome			
Saginaw	1	0	17	16	Sioux City Journal	?	16
Sioux City	1	1	21	22		22	?
Battle Creek	2	0	13	14	Midland Daily News	?	14
Midland	2	1	12	7		7	?
Oxford	3	0	26	23	Lowell Sun	?	23
Lowell	3	1	25	27		27	?
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Small group exercise: Can you think of **more than one way** to learn about the counterfactual causal effect of treatment using what we observe?

What is the true effect of the treatment assignment?



We don't know.



What is the true effect of the treatment assignment?



I don't know the truth, but I can provide a good guess of the average causal effect.

i	z_i	y_i	y_{i1}	y_{i0}
A	0	16	?	16
B	1	22	22	?
C	0	7	?	7
D	1	14	14	?
			$\bar{y}_{i1}$	$\bar{y}_{i0}$

$$\begin{aligned}\widehat{ATE} &= \bar{y}_{i1} | z_i = 1 - \bar{y}_{i1} | z_i = 0 \\ &= \frac{22+14}{2} - \frac{16+7}{2} = 6.5\end{aligned}$$

What is the true effect of the treatment assignment?

I dew nut knew thee truth,
but, given pryors, I cane
predikte itf
probabeeleete .



i	z_i	y_i	y_{i1}	y_{i0}
A	0	16		16
B	1	22	22	
C	0	7		7
D	1	14	14	

$$P(\text{bell curve}, f(y_1 - y_0)) = \text{bell curve}$$

What is the true effect of the treatment assignment?

I don't know the truth,
but I can assess specific
claims about the truth.



$H_0: y_{i1} = y_{i0}$

i	z_i	y_i	y_{i1}	y_{i0}
A	0	16	?	16
B	1	22	22	22
C	0	7	?	7
D	1	14	14	14

$P(t(Y, z))$

$\frac{1}{6}$

$$P = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

-8.5 -6.5 -1.5 .5 (6.5) 8.5 $t(Y, z)$

Testing the Sharp Null of No Effects Practically I

Ingredients: Hypothesis, Test Statistic Function, Distribution

Testing the Sharp Null of No Effects Practically II

Ingredients: Hypothesis, Test Statistic Function, Distribution

```
newexperiment <- function(block){
  block_ra(blocks=block)
}

testStat1 <- function(y,z,b){
  ## Diff of means
  ## y is outcome
  ## z is treatment (binary here)
  ## b is block
  if(length(unique(b))==1){
    ## if only one block, then this is a simple mean difference
    return(mean(y[z==1]) - mean(y[z==0]))
  } else {
    z <- z[order(b)]
    ys <- y[order(b)]
    ysdiffs <- ys[z==1] - ys[z==0] ## assuming sorted by pair
    meandiffs <- mean(ysdiffs)
    return(meandiffs)
  }
}

testStat2 <- function(y,z,b){
  ## Diff of mean ranks
  ## y is outcome
  ## z is treatment (binary here)
  ## b is block
  y <- rank(y) ##
  if(length(unique(b))==1){
    ## if only one block, then this is a simple mean difference
    return(mean(y[z==1]) - mean(y[z==0]))
  } else {
    z <- z[order(b)]
    ys <- y[order(b)]
    ysdiffs <- ys[z==1] - ys[z==0] ## assuming sorted by pair
    meandiffs <- mean(ysdiffs)
    return(meandiffs)
  }
}
```

Testing the Sharp Null of No Effects Practically I

Ingredients: Hypothesis, Test Statistic Function, Distribution

The probability distribution implied by the hypothesis is generated by repeating the experiment.

```
obsTestStat1 <- with(newsd, testStat1(y=r, z=z, b=sF))
obsTestStat2 <- with(newsd, testStat2(y=r, z=z, b=sF))
set.seed(12345)
refDistNull1 <- replicate(1000, with(newsd, testStat1(y=r, z=newexperiment(b=sF), b=sF))
refDistNull2 <- replicate(1000, with(newsd, testStat2(y=r, z=newexperiment(b=sF), b=sF))

p1SideMean <- mean(refDistNull1>=obsTestStat1)
p1SideRank<-mean(refDistNull2>=obsTestStat2)
p2SideMean<-2*min(mean(refDistNull1>=obsTestStat1), mean(refDistNull1<=obsTestStat1))
p2SideRank <- 2*min(mean(refDistNull2>=obsTestStat2), mean(refDistNull2<=obsTestStat2))
p2SideRank
```

[1] 0.712

p2SideMean

[1] 0.754

Mean Difference Test Statistic

$H_0 : y_{i1} = y_{i0}$, One-sided $p = 6/16 = .375$.

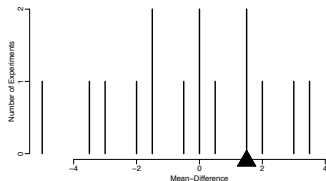


Figure 2: The randomization distribution of the mean-difference test statistic under the null hypothesis of no effects is shown by the tall black lines. The black triangle shows the value of the mean-difference observed in the actually fielded experiment.

```
6/16
```

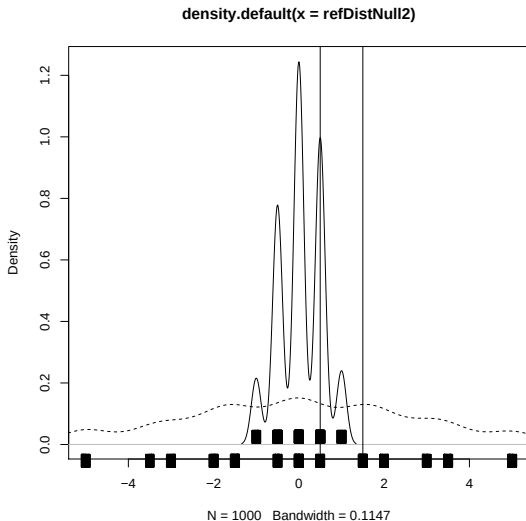
```
[1] 0.375
```

```
p1SideMean
```

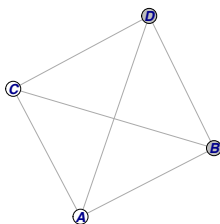
```
[1] 0.377
```

Different Test Statistics Summarize Information Differently

- Power differences. Sensitivity to different alternatives.

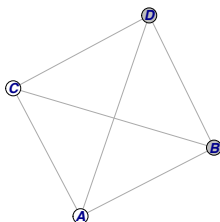


Statistical inference for counterfactual quantities with interference?



i	Z_i	Y_i	$y_{i,1100}$	$y_{i,0101}$	$y_{i,1001}$	$y_{i,0110}$	$y_{i,1010}$	$y_{i,0011}$
A	0	16	?	16	?	?	?	?
B	1	22	?	22	?	?	?	?
C	0	7	?	7	?	?	?	?
D	1	14	?	14	?	?	?	?

Statistical inference for counterfactual quantities with interference?



i	Z_i	Y_i	$y_{i,1100}$	$y_{i,0101}$	$y_{i,1001}$	$y_{i,0110}$	$y_{i,1010}$	$y_{i,0011}$	$y_{i,0000} \equiv y_{i,0}$
A	0	16	?	16	?	?	?	?	16
B	1	22	?	22	?	?	?	?	22
C	0	7	?	7	?	?	?	?	7
D	1	14	?	14	?	?	?	?	14

The sharp null of no effects is a model of no interference:

$H_0 : y_{i,1100} = y_{i,0101} = y_{i,1001} = y_{i,0110} = y_{i,1010} = y_{i,0011} = y_{i,0000},$
 $y_{i,0} = \mathcal{H}(y_{i,z}, \mathbf{0}) = y_{i,z}, p = 0.33.$

Introducing the **uniformity trial** $\equiv y_{i,0000}$ (Rosenbaum, 2007).

Permutations vs Central Limit Theorem?

Which distribution should we use? Which p -value is better? (On what basis would we judge a p -value?)

```
lmpvalue <- summary(lm(r~z+sF,data=newsdf))$coef["z",4]  
lmpvalue  
  
[1] 0.885249  
  
p2SideMean  
  
[1] 0.754
```

- (1) False positive rate
- (2) Power

Permutations vs Central Limit Theorem?: False Positive Rate

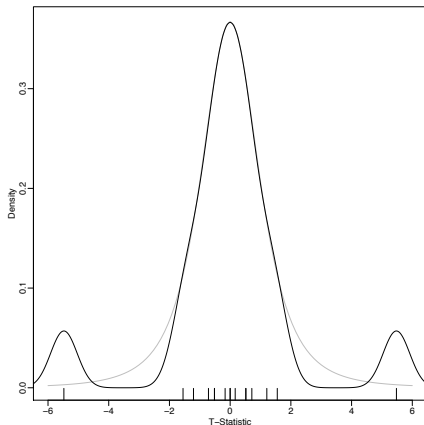


Figure 5: Under the null hypothesis of no effects, t -statistics calculated after repeating the experimental assignment process should be distributed around zero. The exact randomization distribution is tri-modal for the Newspapers study and does not match the t -distribution that we would expect under a larger sample.

Permutations vs Central Limit Theorem?: False Positive Rate

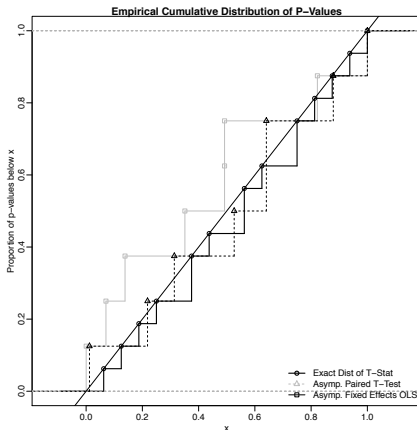


Figure 6: Cumulative distributions of p-values for testing the null hypothesis of no effects. A valid test would be at or below the diagonal line. Tests which would encourage rejection too quickly have points above the line.

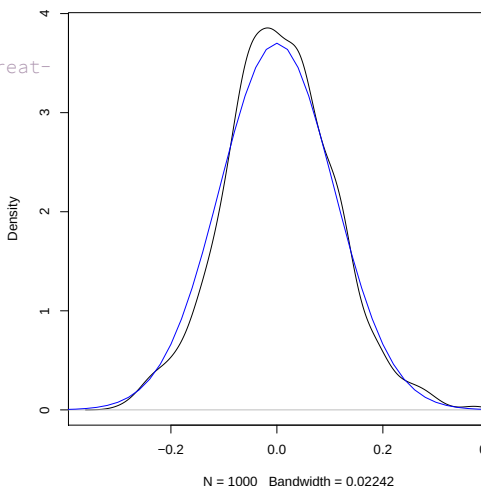
What about when samples are large?

```
set.seed(20180225)
N <- 1000
y0 <- rchisq(N,df=1)
y1 <- y0 + rnorm(N,mean=0,sd=sd(y0))## No treatment effect but different variance
Z <- complete_ra(N) ## very simple randomization
Y <- Z*y1 + (1-Z) * y0 ## randomization reveals a potential outcome

newexpComplete <- function(N){
  complete_ra(N)
}

testStat3 <- function(y,z){
  ## Diff of means
  ## y is outcome
  ## z is treatment (binary here)
  md <- mean(y[z==1]) - mean(y[z==0])
  return(md)
}
```

density.default(x = randDist)



What does it mean to test a hypothesis?

What is a p -value?

What do we learn from a hypothesis test?

What do we need to assume to make a hypothesis test (in this simplest case) interpretable?

Why test a hypothesis? What is a hypothesis?

- (1) In an experiment a hypothesis is a claim or model about the relationship among potential outcomes; about unobserved outcomes.
- (2) In an experiment a hypothesis test allows a form of causal inference (or statistical inference about a counterfactual causal effect)

Where does the reference distribution come from? What justifies it?

How to make a distribution? Where do they come from? How do we justify our choices?

In an experiment we have the opportunity to use randomization to provide a "reasoned basis for inference". That is, we know how randomization happened. And the specific randomization could have happened differently (different units could have been in treatment and control depending on the random process).

Key features of Fisher's approach

Flexible Any scientific model than can generate implications for all units' potential outcomes can, in principle, produce testable parameters.

Design based Requires knowledge of probability of Z not Y or $Y|X$ or $\beta|\gamma$.

Finite Sample Oriented Does not require asymptopia. Can use asymptopia when there for a visit.

Can be slow In between 8 cities and asymptopia is a land of many permutations.

Probably conservative Uses relatively little of the total information we have available about the science.

If you want to know more read Paul Rosenbaum's work The version of Fisher's approach I discuss here is built on work by Paul Rosenbaum. Read his work if you want to learn more.

Panagopoulos, Costas. 2006. "The Impact of Newspaper Advertising on Voter Turnout: Evidence from a Field Experiment." Paper presented at the MPSA 2006.

What range of effects might be surprising?

Hypothesize a model of potential outcomes For $H_0 : y_{i1} = y_{i0} + \tau$, what τ might be surprising?

Map the model to observation via design What would $\tau = 6$ imply for what we observe? If $\tau = 6$ and $Y_i = Z_i y_{i1} + (1 - Z_i) y_{i0}$ then $Y_i - Z_i 6 = y_{i0}$.

Generate the randomization distribution of this hypothesis As before

Summarize information against the hypothesis For hypotheses of $\tau \geq 6$ we have $p \leq .125$ using a mean difference test statistic.

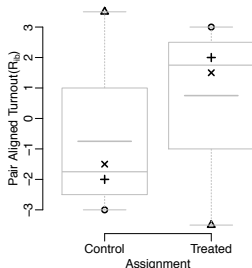


Figure 3: The distribution of outcomes in the control and treatment groups. The values of turnout are “pair aligned” or “pair mean centered” so that we can focus attention on the paired differences rather than on the differences in turnout levels between pairs.

What effects might be least surprising?

The idea of a “best guess” maps onto the Hodges-Lehmann point estimate: the hypotheses for which $E(t(Z, Y)) = 0$: ex. the difference of means is zero. Here $\tau = 1.5$ for the mean difference and $\tau = 3.25$ for the rank-based test (which equalizes medians).

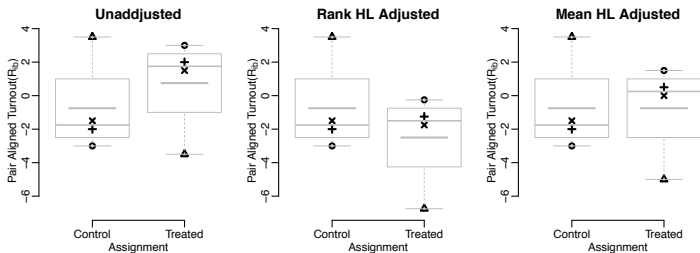


Figure 4: The unadjusted comparison of controls versus treated (left most plot) as compared to the results of adjusting the treatment group based on the Hodges-Lehmann (HL) point estimates derived from two different test statistics (Wilcoxon paired signed ranks and the mean difference). Notice that the control group remains the same and it is the hypotheses which imply changes in the distribution of the treatment group compared to the unadjusted, observed, outcomes.

Theoretical models of potential outcomes can produce sharp hypotheses

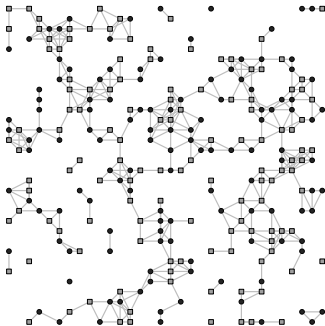


Figure: A simulated data set with 256 units and 512 connections. The $256/2 = 128$ treated units ($Z_i = 1$) are shown as filled circles and an equal number of control units ($Z_i = 0$) are shown as as gray squares.

Theoretical models of potential outcomes can produce sharp hypotheses

$$\mathcal{H}(\mathbf{y}_0, \mathbf{z}, \beta, \tau) = \left[\beta + (1 - z_i)(1 - \beta) \exp \left(-\tau^2 \mathbf{z}^T \mathbf{S} \right) \right] \mathbf{y}_0 \quad (1)$$

$$\mathcal{H}(\mathbf{y}_z, \mathbf{0}, \beta, \tau) = \left[\beta + (1 - z_i)(1 - \beta) \exp \left(-\tau^2 \mathbf{z}^T \mathbf{S} \right) \right]^{-1} \mathbf{y}_z \equiv \mathbf{y}_0 \quad (2)$$

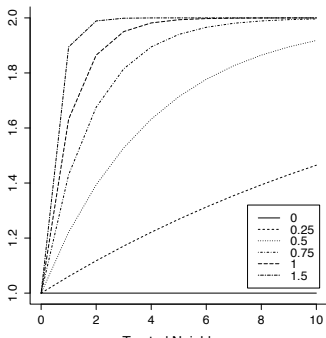


Figure: Growth curve of spillover effects for the expression $\beta + (1 - \beta) \exp \left(-\tau^2 \mathbf{z}^T \mathbf{S} \right)$ as the number of treated neighbors, $\mathbf{z}^T \mathbf{S}$, increases for $\beta = 2$ and a selection of τ values.

Theoretical models of potential outcomes can produce sharp hypotheses

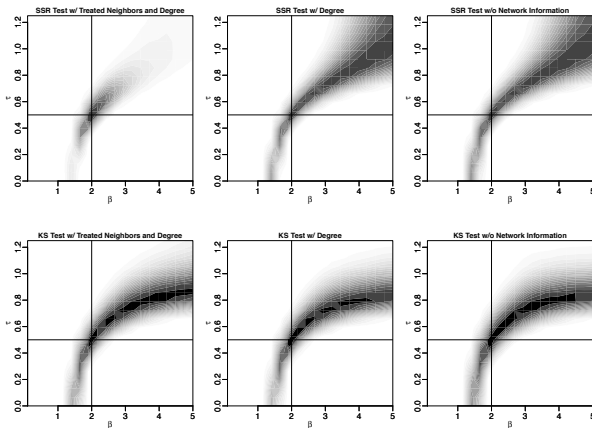


Figure: Proportion of p -values less than .05 for randomization tests of joint hypotheses about τ and β . Darker values mean less rejection. Truth is at $\tau = .5, \beta = 2$. All tests reject the truth no more than 5% of the time at $\alpha = .05$. All simulations, not Normal approximations.

A General Fisherian Inference Algorithm

- 1 Write a model $(\mathcal{H}(\mathbf{y}_0, \mathbf{z}, \theta))$ converting uniformity trial into observed data (i.e. a causal model).
- 2 Solve for $\mathbf{y}_0$: $\mathcal{H}(\mathbf{y}_z, \mathbf{0}, \theta_0) = \mathbf{y}_0$
- 3 Select a test statistic that is effect increasing in all relevant dimensions.
- 4 Compute p -values for substantively meaningful range of θ . Or calculate boundaries of regions.

Robust test statistics can increase power.

For the simple mean difference test statistic, we have $p = .375$, for a rank sum test (the sum of the ranks of the treated units), we have a $p = .4375$, and for an M-estimator based test (like mean-differences but with weights roughly inversely proportional to the Cook's D influence measure) we have $p = .3125$.

Cook's D	MM-Weights	City	Pair	Treatment	Turnout
0.13	1.00	Saginaw	1	0	16
0.13	1.00	Sioux City	1	1	22
0.48	0.94	Battle Creek	2	0	14
0.48	0.94	Midland	2	1	7
0.04	1.00	Oxford	3	0	23
0.04	1.00	Lowell	3	1	27
0.01	1.00	Yakima	4	0	58
0.01	1.00	Richland	4	1	61

Table 2: Not all cases have the same influence on the mean difference. Cook's D summarizes this influence — Battle Creek and Midland have disproportionate influence. A robust fit (details in the text) downweights Battle Creek and Midland.

Covariance adjusted tests can increase power.

Using the difference pre-vs-post as the outcome (comparing treated pre-vs-post with paired control pre-vs-post), and using the robust test statistic for $H_0 : y_{i1} = y_{i0}$ $p = 4/16 = .25$.
Using $e_i = (Y_i - Y_{i,t-1}) - (\hat{\beta}_0 + \hat{\beta}_1 \text{pop} + \hat{\beta}_2 \text{num candidates})$ and the robust test statistic to test $H_0 : y_{i1} = y_{i0}$ we have $p = 2/16 = .125$.